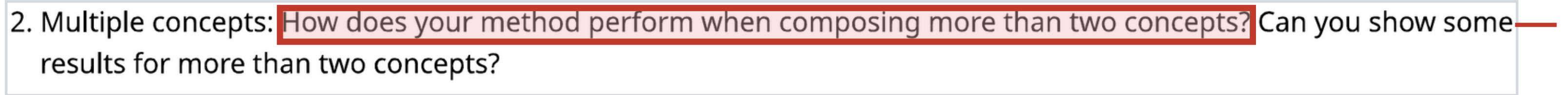
G10: Beyond two animals on SDXL

PrzE question 2, UHz6 weakness 3 (scope) · status: partly answered (SDXL only)

THE REVIEWER'S POINT

PrzE, question 2



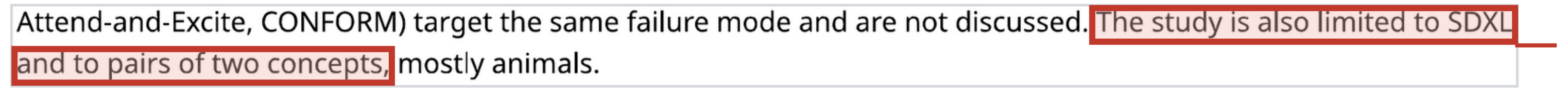
2. Multiple concepts: How does your method perform when composing more than two concepts? Can you show some results for more than two concepts?

P-Q2

pairs only

Reviews.pdf, page 2 (no text layer; passage located by OCR)

UHz6, weakness 3 (scope)



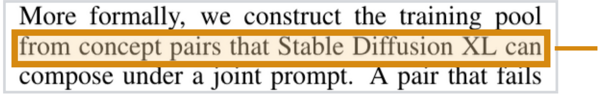
Attend-and-Excite, CONFORM) target the same failure mode and are not discussed. The study is also limited to SDXL and to pairs of two concepts, mostly animals.

U-W3c

Reviews.pdf, page 3 (no text layer; passage located by OCR)

THE SUBMITTED WEAKNESS

Section 4, the training pool



More formally, we construct the training pool from concept pairs that Stable Diffusion XL can compose under a joint prompt. A pair that fails

pairs on SDXL only

56173_Restoring_plurality_in_c.pdf (submitted), page 7

three concepts

discussed

still SDXL

THE REVISED ANSWER

Figure 13, three concepts

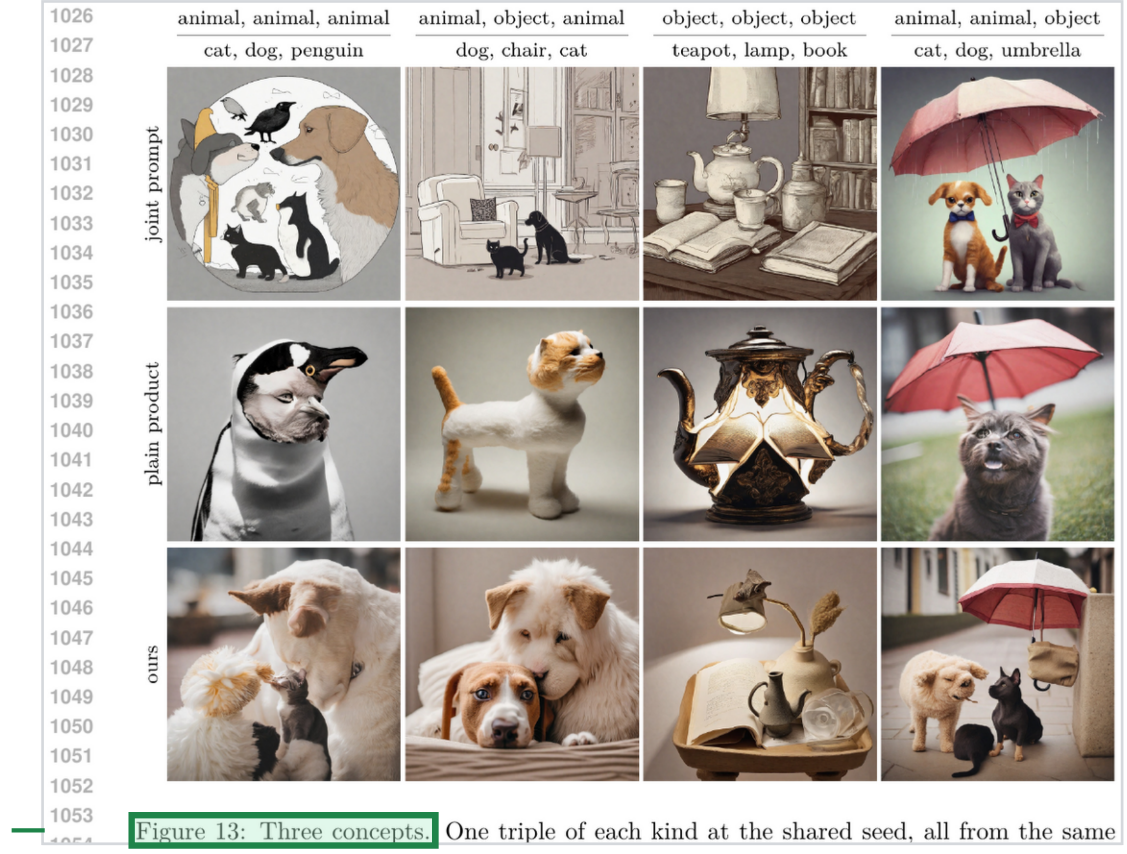
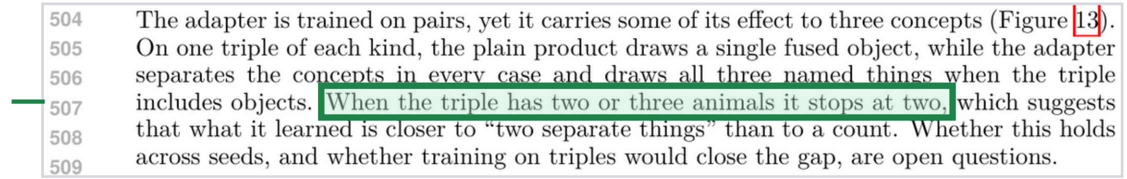


Figure 13: Three concepts. One triple of each kind at the shared seed, all from the same

one triple of each kind

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 20

Discussion, three concepts



The adapter is trained on pairs, yet it carries some of its effect to three concepts (Figure 13). On one triple of each kind, the plain product draws a single fused object, while the adapter separates the concepts in every case and draws all three named things when the triple includes objects. When the triple has two or three animals it stops at two, which suggests that what it learned is closer to "two separate things" than to a count. Whether this holds across seeds, and whether training on triples would close the gap, are open questions.

stops at two animals

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 10

Not in the revision

No model other than SDXL. The SD 1.5 results were removed on 2026-11-12.

no crop: this panel states what the claim table records

The reply we plan to send (replies/przе.md: Q2)

partly answered

Q2: multiple concepts

Appendix C.5 (Figure 13) shows one triple of each kind. The plain product fuses the three concepts or drops one; the adapter, trained only on pairs, separates them in every case and draws all three named things when the triple includes objects, but with two or three animals it draws only two. The Discussion now addresses this.

Objects beyond animals are scored in Table 1 (animals with objects, objects); see G4. UHz6's weakness 3 reply is on the G5 page. Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.